Tharwat Farag Badreldin

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LinkedIn | GitHub

Professional Summary

Microsoft-certified professional holding DP-700, DP-600, DP-800, and DP-900 certifications, with 4+ years of hands-on experience in SQL, relational databases, database administration, and production troubleshooting. Completed the Digilians Data Engineering Program at the Egyptian Military Academy in collaboration with the Ministry of Communications and Information Technology (MCIT), gaining practical experience in Python, ETL/ELT pipelines, data modeling, Microsoft Fabric, and Azure. Previously led the E-learning/EdTech sector at Camp Coding, overseeing 100+ educational platforms serving more than 130,000 active users.

Education

B.Sc. in Computer Science
Sep 2019 – Jun 2023

Faculty of Computers and Information, Tanta University — *Tanta, Egypt*

- Graduated with a Very Good grade.

Work Experience

Data Engineering Specialist Trainee
Jan 2026 – Jun 2026

Digilians – Egyptian Military Academy, in collaboration with MCIT — *Cairo, Egypt*

- Applied prior SQL and database experience to develop ETL/ELT workflows using Python and SQL.
- Practiced SQL query optimization, indexing strategies, and relational data modeling for analytical workloads.
- Performed data extraction, transformation, validation, and loading to prepare structured datasets for analytics and reporting.

Technical Lead – LMS & EdTech Platforms
Jul 2020 – Dec 2025

Camp Coding — *Tanta, Egypt*

- Led the company's E-learning/EdTech operations, managing client requirements, platform delivery, technical support, and ongoing client relationships.
- Oversaw the development and deployment of 100+ educational/LMS platforms across Egypt and the Middle East, serving more than 130,000 active users in production.
- Resolved production data issues at the SQL/database level through data validation, record consistency checks, database troubleshooting, root-cause analysis, and database administration.
- Led technical team coordination and investigated application-level and backend configuration issues, handling technical escalations from clients and internal teams.

IT Manager
Nov 2023 – Jul 2025

Baseeta — *Part-time*

- Oversaw IT operations with a focus on system reliability and operational efficiency.
- Led website optimization initiatives focused on performance and security, aligning technical work with business requirements.

Software Sales Specialist
Dec 2024 – Jun 2025

Alnaierh — *Riyadh, Saudi Arabia – Remote*

- Supported software-related operations and coordinated with technical teams to address operational and website performance requirements.
- Facilitated communication between technical teams and business stakeholders to support software delivery and client needs.

Selected Data Engineering Projects

B.Laban Retail Customer & Sales Analytics Platform

GitHub

- Built an end-to-end retail data pipeline using PySpark, Microsoft Fabric, Delta Lake, and SQL, implementing a Bronze → Silver → Gold medallion architecture.
- Implemented data-quality validation, standardization, anomaly flagging, and automated tests across 7K transactional records, identifying 48 invalid transactions while preserving source-data traceability.
- Prepared Gold datasets for branch, product, channel, customer, and daily-sales analytics to support downstream reporting.

RetailFlow Supply Chain & Inventory Data Engineering Pipeline

GitHub

- Built a supply chain data pipeline using Python, PostgreSQL, and Docker, transforming raw data through Bronze and Silver layers into a dimensional analytics warehouse.
- Implemented schema contracts, data-quality validation, deterministic deduplication, quarantine handling, and SHA-256 ingestion manifests for source traceability.
- Designed a Star Schema for inventory snapshots and purchase-order fulfillment, with transactional, idempotent loading and a documented 55-test automated suite.

SSIS Data Warehouse ETL Pipeline

GitHub

- Built a SQL Server data warehouse using SSIS and Star Schema modeling, implementing ETL workflows with SCD Type 1/2 handling and SSAS/SSRS reporting.

Apache Airflow ETL Pipeline

GitHub

- Developed and orchestrated automated ETL workflows using Apache Airflow and Astronomer in a Dockerized environment.

Real Estate Data Platform

GitHub

- Developed a Python and Pandas pipeline to process web-scraped property listings, using SQL Server for data storage and analytical SQL views for sale and rental KPIs.
- Implemented data cleaning and quality checks to produce structured, analysis-ready datasets.

Technical Skills

Programming & Query Languages: Python, SQL, JavaScript

Databases: SQL Server, PostgreSQL, MySQL, SQLite

SQL & Database Engineering: SQL Development, Database Administration, Database Troubleshooting, Data Validation, Data Integrity, Query Optimization, Indexing, Database Deployment, Database Security

Data Engineering: ETL/ELT Pipelines, Data Extraction, Data Transformation, Data Loading, Data Warehousing, Data Modeling, Relational Data Models, Star Schema, Apache Airflow, SSIS, Docker

Microsoft Fabric: Data Factory, Data Pipelines, Lakehouse, OneLake, Data Warehouse, Notebooks, Semantic Models

Big Data & Cloud: Apache Spark, Apache Kafka, Hadoop, Hive, Microsoft Azure

Data Processing & Tools: Pandas, NumPy, Selenium, Git, GitHub, Jupyter Notebook, Visual Studio Code

Analytics & Reporting: Business Intelligence, SSAS, SSRS, Analytical SQL

Leadership: Technical Leadership, Team Leadership, Client Management, Stakeholder Management, Requirements Gathering, Cross-functional Collaboration

Certifications

Microsoft Certified: Fabric Data Engineer Associate (DP-700)

Microsoft Certified: Fabric Analytics Engineer Associate (DP-600)

Microsoft Certified: SQL AI Developer Associate (DP-800)

Microsoft Certified: Azure Data Fundamentals (DP-900)

Languages

Arabic: Native

English: Professional Working Proficiency